The permanent of the GCD matrix

Vico Bonfioli*

July 2026

Abstract

Let $a(n) = \mathrm{per}[\gcd(i,j)]_{1\le i,j\le n}$. Whereas the determinant $\det[\gcd(i,j)] = \prod_{k\le n}\varphi(k)$ has been known since Smith (1876), the permanent—sequence `A085244` in the OEIS—appears not to have been studied beyond its definition and values. We compute $a(n)$ for $n \le 30$, prove a family of congruences—$p \mid a(n)$ once there are at least p integers $\le n$ all of whose prime factors are $\equiv 1 \pmod p$ (giving $2 \mid a(n)$ for $n \ge 3$, sharpened to $4 \mid a(n)$ for $n \ge 4$, and $3 \mid a(n)$ for $n \ge 13$), and show $v_2(a(n)) \to \infty$ via a $(\mathbb{Z}/2)^k$ group action on the "$\equiv 1 \bmod 2^k$" rows—and prove $(a(n)/n!)^{1/n} \to \infty$: writing $(a(n)/n!)^{1/n} = (\log n)^{\theta+o(1)}$, the exponent obeys $2\ln\varphi - \varphi^{-2} \le \theta \le \log 2$ (numerically $\theta \approx 0.68$; whether $\theta = \log 2$ or a strictly smaller constant is open). The upper bound $\theta \le \log 2$ is elementary; the lower bound is a doubly stochastic construction from prime-divisibility blocks (via Van der Waerden–Falikman–Egorychev) whose per-prime gains g_p satisfy $p\,g_p \to 2\ln\varphi - \varphi^{-2}$ (with φ the golden ratio), so $\sum_p g_p = \infty$ by Mertens [6]—giving $c_\infty = \infty$ unconditionally. Finally, we explain the 2-adic *deficit* $v_2(n!) - v_2(a(n))$ at $n = 2^k + 1$ through a mechanism rather than opaque cancellation: two exact valuation theorems for odd permanents (of size 2^a, resp. $2^a + 1$, driven by the first, resp. second, permanental symmetric function) combine to give $D_{N_0}(2^k+1) = 2k-4$ — for the *weight-zero grade* N_0 of a, not for a itself — whenever a computable, permanent-free "achiever parity" is odd, which occurs for 11 of the 20 values $k \le 23$. So that grade's deficit is $\Theta(\log n)$ conditional on the parity's infinitude, one open step; transferring the conclusion to a itself needs $v_2(a) = v_2(N_0)$, which we verify only at $k \le 5$ and where the grade margins are small, so we do not claim it in general. Reducing that parity through a chain of verified identities (machine-checked at five links) yields the following description: the boundary contribution of each prime $p \equiv 3$ (4) is exactly the *binary expansion of* $1/p$, and the resulting words' statistics form a class-number ladder of *depth exactly three*—density, digit-pair, and digit-triple counts are exact formulas in $h(-p)$ and $h(-8p)$ (e.g. the number of 11-pairs per period is $(p - 4h(-p) - 3)/8$), while digit-quadruple and finer statistics provably admit no class-number formula, the ladder terminating because no real primitive Dirichlet character of conductor 16 exists. The 2-adic deficit of the gcd permanent is thus completely stratified: a finite class-number core over $\mathbb{Q}(\sqrt{-p})$ and $\mathbb{Q}(\sqrt{-2p})$, and a character-sum (Pólya–Vinogradov) tail.

1 The sequence

$a(n) = \sum_{\sigma\in S_n}\prod_{i=1}^{n}\gcd(i,\sigma(i))$ begins $1, 3, 14, 112, 872, 14372, 154480, \ldots$ (30 terms computed exactly, by a Gray-code Ryser evaluation [4] and, for $n > 27$, a modular Ryser with CRT).

*Data and scripts: `code/` in the `rama-notes` repository; an independent Lean/Mathlib formalization of Theorem 2's engine is in `RamaLean/Paper3Permanent.lean`. Correspondence: `vicobonfioli@gmail.com`.

It is sequence A085244 in the OEIS (Dekel, 2003, with a b-file to $n = 38$); its arithmetic—the subject of this note—appears not to have been studied, and the derived 2-adic valuation, deficit, and achiever-parity sequences are new. Computing permanents is $\#P$-hard (Valiant), and—unlike the determinant—there is no product formula.

2 Congruences

For an arithmetic function the determinant $[\,f(\gcd(i,j))\,]$ has Smith's product form; the permanent has no such structure, but retains congruences.

Proposition 1. *$a(n) \equiv \prod_{k\le n} \varphi(k) \pmod 2$; in particular $2 \mid a(n)$ for $n \ge 3$.*

Proof. Over $\mathbb{F}_2$ permanent and determinant agree, so $a(n) \equiv \det[\gcd(i,j)] = \prod_{k\le n}\varphi(k)$; the factor $\varphi(3) = 2$ makes this even for $n \ge 3$. □

Theorem 2. *Let p be prime and $A_p(n) = \#\{j \le n : \text{every prime factor of } j \equiv 1 \pmod p\}$. If $A_p(n) \ge p$ then $p \mid a(n)$. Consequently $2 \mid a(n)$ for $n \ge 3$ and* **$3 \mid a(n)$ *for all*** *$n \ge 13$ (the third such j is 13); more generally $5 \mid a(n)$ for $n \ge 61$, etc.*

Proof. Lemma A. If a matrix B over $\mathbb{Z}$ has p columns pairwise congruent mod p, then $\mathrm{per}(B) \equiv 0 \pmod p$. Indeed, let T index p such columns (common value v mod p); group the permutations by the pair (R,ρ) with $R = \sigma^{-1}(T)$ and $\rho = \sigma|_{[n]\setminus R}$. Exactly $p!$ permutations give each (R,ρ), and for all of them $\prod_i B_{i,\sigma(i)} \equiv (\prod_{i\in R} v_i)\prod_{i\notin R} B_{i,\rho(i)} \pmod p$; summing, $\mathrm{per}(B) \equiv p!\sum(\cdots) \equiv 0$.

Lemma B. As i runs over $[1,n]$, $\gcd(i,j)$ takes every divisor of j; so the j-th column of $[\gcd(i,j)]$ is $\equiv (1,\ldots,1) \pmod p$ iff every divisor—equivalently every prime factor—of j is $\equiv 1 \pmod p$.

If $A_p(n) \ge p$ there are $\ge p$ such columns, all $\equiv (1,\ldots,1) \bmod p$; apply Lemma A. The counts follow from the $p = 2$ (j odd) and $p = 3$ ($j \in \{1,7,13,\ldots\}$) cases. □

Theorem 2 is formalized *end to end* in Lean 4/Mathlib with only the three standard axioms (no `sorry`, no `native_decide`): `three_dvd_gcd_permanent` proves $3 \mid a(n)$ for all $n \ge 13$ over $\mathbb{Z}/3$, via Lemma A (`permanent_eq_zero_of_col_period`, over `Matrix.permanent`) and the 3-cycle through columns $1, 7, 13$. The $p = 2$ case can be sharpened past $2 \mid a(n)$:

Proposition 3. *$4 \mid a(n)$ for all $n \ge 4$.*

Proof. By the permanent–determinant identity $a(n) = \det(M) + 2\Sigma$, where $\Sigma = \sum_{\sigma\text{ odd}}\prod_i \gcd(i,\sigma i)$. Smith's theorem gives $\det(M) = \prod_{k\le n}\varphi(k)$, and $\varphi(3)\varphi(4) = 4$ makes $4 \mid \det(M)$ for $n \ge 4$. Modulo 2, $\Sigma \equiv \#\{\sigma \text{ odd} : \prod_i \gcd(i,\sigma i) \text{ odd}\}$. A product $\prod \gcd(i,\sigma i)$ is odd iff no index has both $i, \sigma(i)$ even, i.e. σ carries the $\lfloor n/2\rfloor$ evens into the $\lceil n/2\rceil$ odds; there are exactly $(\lceil n/2\rceil!)^2$ such σ (choose an injection evens$\hookrightarrow$odds, $\lceil n/2\rceil!/(\lceil n/2\rceil - \lfloor n/2\rfloor)!$ ways, then a bijection of the rest, $\lceil n/2\rceil!$ ways). The indicator matrix $G = [\gcd(i,j) \text{ odd}] = J - ee^\top$ (e = evens) has rank ≤ 2, so $\det G = 0$ for $n \ge 3$; hence the odd and even such σ are equinumerous and $\#\{\sigma \text{ odd} : \prod \text{ odd}\} = \frac{1}{2}(\lceil n/2\rceil!)^2$, even for $n \ge 3$. Thus $2\Sigma \equiv 0 \pmod 4$, and $4 \mid a(n)$ for $n \ge 4$. □

Proposition 3 is formalized *end to end* in Lean 4/Mathlib (`RamaLean/Paper3FourDivides.lean`, standard axioms only, no `sorry`): the chain includes Smith's determinant $\det[\gcd(i,j)] = \prod_{k\le n}\varphi(k)$ (via the factorization $M = LDL^\top$) and the involution argument for $2 \mid \Sigma$ over $\mathbb{Z}/2$.

Theorem 4 (the 2-adic tower)**.** *$v_2(a(n)) \to \infty$. Explicitly $v_2(a(n)) \ge k+1$ whenever $[1,n]$ contains 2^k integers all of whose prime factors are $\equiv 1 \pmod{2^k}$; this holds for all large n by Dirichlet, giving $v_2(a(n)) \ge (\frac{1}{2} - o(1))\log_2 n$.*

Proof. Fix $k \ge 2$. If every prime factor of m is $\equiv 1 \pmod{2^k}$ then so is every divisor of m, so $\gcd(m,\cdot) \equiv 1 \pmod{2^k}$: the row of $[\gcd(i,j)]$ at such an index m is $\equiv (1,\dots,1) \pmod{2^k}$. By Dirichlet there are infinitely many primes $\equiv 1 \pmod{2^k}$; for n large pick 2^k such row-indices $R \subseteq [n]$, identify R with $(\mathbb{Z}/2)^k$, and let $G \le S_n$ be its regular representation (permuting R, fixing $[n]\setminus R$). Then $|G| = 2^k$ and each nontrivial $g \in G$ is a product of 2^{k-1} disjoint transpositions, so $\operatorname{sgn}(g) = (-1)^{2^{k-1}} = 1$ (as $k \ge 2$), i.e. $G \le A_n$. Writing $a(n) = \det + 2\Sigma$ with $\Sigma = \sum_{\operatorname{sgn}\sigma=-1}\prod_i \gcd(\sigma i, i)$, the group G acts on $\{\sigma : \operatorname{sgn}\sigma = -1\}$ by $\sigma \mapsto g\sigma$ (sign-preserving since $G \le A_n$; free, being a group acting by translation). As g permutes R and fixes the rest, $\prod_i \gcd(g\sigma i, i) \equiv \prod_i \gcd(\sigma i, i) \pmod{2^k}$, so mod 2^k the summand is constant on each G-orbit and each orbit (of size $|G| = 2^k$) sums to 0; hence $2^k \mid \Sigma$. Since $2^{k+1} \mid \det = \prod_{j\le n}\varphi(j)$ for large n ($v_2(\det) = \sum_{j\le n} v_2\varphi(j) \to \infty$), we get $2^{k+1} \mid a(n)$, i.e. $v_2(a(n)) \ge k+1$. The rate follows since the count of admissible $m \le n$ is $\ge \pi(n; 2^k, 1) \sim n/(2^{k-1}\log n)$, so k may grow to $\sim \frac{1}{2}\log_2 n$. □

Remark 5. This settles the tower ($k = 1$, i.e. $2 \mid a(n)$, is the involution $(0\,2)(1\,3)$ of Proposition 3; $8 \mid a(n)$ for $n \ge 17$ uses the Klein four-group on rows $m = 1, 5, 13, 17$, all $\equiv 1 \bmod 4$). The case $8 \mid a(n)$ is *formalized end to end in Lean 4/Mathlib* for $n \ge 17$ (theorem `Paper3Four.eight_dvd_permanent`, three standard axioms, no `sorry`): the regular Klein-four $V_4 = \langle (c_0c_1)(c_2c_3), (c_0c_2)(c_1c_3)\rangle$ acts on the four rows $m = 1, 5, 13, 17$; freeness is immediate (every nontrivial element moves c_0), and the non-disjoint commutation is supplied by the conjugation lemma $e(c\,d)e^{-1} = (ec)(ed)$. It reuses the orbit-sum engine `sum_zmod_eq_zero_of_free_invariant`. Empirically the true rate is linear, $v_2(a(n)) \approx v_2(n!) \sim n$ (with $v_2(n!) - v_2(a(n)) \in [-1, 6]$ through $n = 34$, by a mod-2^{128} Ryser)—far above the $\frac{1}{2}\log_2 n$ the construction guarantees, since most terms $\prod_i \gcd(\sigma i, i)$ carry high 2-adic valuation while the group action only extracts the "$\equiv 1 \bmod 2^k$" rows. A *linear* lower rate $v_2(a(n)) \gtrsim n/2$ is proved unconditionally in Theorem 6; this is sharpened to the constant $c = 1$, $v_2(a(n)) \ge n - 2\lfloor \log_2 n\rfloor - 2$, in Theorem 9. The deficit $v_2(n!) - v_2(a(n))$ appears to be *bounded*: it reaches 6 at $n = 33$ but does not continue to drift—a factorization $N_0 = \sum_v \operatorname{per}(A_v)\operatorname{per}(B_v)$ of the dominant 2-adic grade makes $v_2(N_0(n))$ computable at $n = 65$, giving $v_2(N_0(65)) = 58$ (deficit 5), which breaks the apparent $2, 4, 6, \dots$ growth along $n = 2^k + 1$. So conjecturally $v_2(a(n)) = v_2(n!) - O(1)$, an even sharper statement than Theorem 9; a rigorous constant bound remains open.

Theorem 6 (linear 2-adic rate)**.** *For all $n \ge 1$,*

$$v_2\big(a(n)\big) \;\ge\; \left\lceil \tfrac{n}{2} \right\rceil - \lfloor \log_2 n \rfloor - 1 \;=\; \tfrac{1}{2}n - O(\log n),$$

so $v_2(a(n)) \to \infty$ linearly.

Proof. By Smith's identity $\gcd(i,j) = \sum_{d|i,\ d|j} \varphi(d)$, expand the permanent one column at a time:

$$a(n) = \sum_{\substack{d=(d_1,\dots,d_n)\\ d_i|i\ \forall i}} \Big(\prod_{i=1}^{n} \varphi(d_i)\Big)\ \mathrm{per}(M_d), \qquad M_d[k,i] = \mathbf{1}[\,d_i \mid k\,],$$

where $\mathrm{per}(M_d) = \#\{\sigma \in S_n :\ d_i \mid \sigma(i)\ \forall i\}$. Fix a tuple d and put $c_1 = \#\{i : d_i = 1\}$, $c_2 = \#\{i : d_i = 2\}$, $c_3 = \#\{i : d_i \geq 3\}$. The c_1 columns with $d_i = 1$ are all-ones, so in any admissible σ they take, in any order, the c_1 values left free by the other columns; hence $\mathrm{per}(M_d) = c_1! \cdot N$ for some integer N, and $v_2(\mathrm{per}(M_d)) \geq v_2(c_1!) = c_1 - s_2(c_1)$ (where s_2 is the binary digit sum). Since $\varphi(d)$ is even for every $d \geq 3$, we have $v_2\big(\prod_i \varphi(d_i)\big) \geq c_3$. Finally $d_i = 2$ forces $2 \mid i$, so $c_2 \leq \lfloor n/2 \rfloor$ and $c_1 + c_3 = n - c_2 \geq \lceil n/2 \rceil$. Thus every term satisfies

$$v_2\Big(\textstyle\prod_i \varphi(d_i)\ \mathrm{per}(M_d)\Big) \ \geq\ (c_1 - s_2(c_1)) + c_3 \ =\ (c_1 + c_3) - s_2(c_1)\ \geq\ \left\lceil \tfrac{n}{2} \right\rceil - s_2(c_1)\ \geq\ \left\lceil \tfrac{n}{2} \right\rceil - \lfloor \log_2 n \rfloor - 1,$$

using $s_2(c_1) \leq \lfloor \log_2 c_1 \rfloor + 1 \leq \lfloor \log_2 n \rfloor + 1$. As $a(n)$ is an integer combination of these terms, $v_2(a(n))$ is at least this common bound. □

Remark 7. The method is tight in *shape* (linear) but not in constant: numerically $v_2(a(n)) \approx v_2(n!) \sim n$ (slope 1; deficit $v_2(n!) - v_2(a(n)) \in [-1, 6]$ through $n = 34$, and $D(2^k + 1) = 2k - 4$ at the peaks $n = 2^k + 1$ for $k \leq 5$, with the corresponding statement for the weight-zero grade proved at every k of odd achiever parity; see Section 4), and the stronger $v_2(a(n)) \geq v_2(\lceil n/2 \rceil!)$ holds for every $n \leq 20$. The factor-2 loss comes from using only the $\lceil n/2 \rceil$ odd positions' freedom; the even positions carry a comparable 2-adic contribution that a single application of the all-ones lemma does not see.

Remark 8 (formalization). Theorem 6 is *formalized end to end in Lean 4/Mathlib* (`RamaLean/Paper3LinearRate.lean`, theorem `two_pow_dvd_permanent`: $2^{\lceil n/2 \rceil - \lfloor \log_2 n \rfloor - 1} \mid a(n)$, three standard axioms, no `sorry`)—the *same* constant $c = \frac{1}{2}$ proved above. The Lean proof formalizes the permanent's Smith expansion $a(n) = \sum_x \big(\prod_i \varphi(x_i)\big)\,\mathrm{per}(M_x)$ (column multilinearity via `Finset.prod_univ_sum`), the all-ones-column divisibility $c_1! \mid \mathrm{per}(M_x)$ (a free `Perm(Fin c1)` action, from the reusable engine `card_dvd_sum_of_free_invariant`: a free finite-group action with invariant summand makes $|G|$ divide the sum over any commutative ring), the *exact* factorial valuation $v_2(c_1!) = c_1 - s_2(c_1)$ (`Nat.sub_one_mul_factorization_factorial`), and the 2-adic bookkeeping.

Theorem 9 (the growth constant: lower bound $c = 1$)**.** *For all* $n \geq 1$, $\ v_2\big(a(n)\big) \ \geq\ n - 2\lfloor \log_2 n \rfloor - 2$, *hence* $\liminf_n v_2(a(n))/n \geq 1$.

Proof. Write $\gcd(i,j) = 2^{\min(v_2 i,\, v_2 j)} h(i,j)$ with $h(i,j) = \gcd(\mathrm{odd}(i), \mathrm{odd}(j))$ *odd*, so $a(n) = \sum_\sigma 2^{w(\sigma)} \prod_i h(\sigma i, i)$ with $w(\sigma) = \sum_i \min(v_2(\sigma i), v_2(i))$. Only pairs with both indices even contribute to w (an odd index has $v_2 = 0$). Grouping the permutations by their even→even sub-injection ρ (an even position sent to an even value),

$$a(n) = \sum_\rho 2^{w_\rho} \Big(\prod_{i \in \mathrm{dom}\,\rho} h(\rho i, i) \Big) \mathrm{Rest}_\rho, \qquad w_\rho = \sum_{i \in \mathrm{dom}\,\rho} \min(v_2(\rho i), v_2(i)) \ \geq\ |\rho|,$$

where $\mathrm{Rest}_\rho = \mathrm{per}(H_\rho)$ and H_ρ is the $(n - |\rho|) \times (n - |\rho|)$ matrix equal to h off the (remaining even)×(remaining even) corner and 0 on that corner (remaining even positions cannot reuse even values). As $\prod h$ is odd, $v_2(\mathrm{term}_\rho) = w_\rho + v_2(\mathrm{Rest}_\rho)$.

Zeroed-corner lemma: if M is $s \times s$, odd off a $b \times b$ corner and 0 on it, then $v_2(\operatorname{per} M) \geq s - 2\lfloor \log_2 s \rfloor - 2$. Indeed the b corner rows are forced into the $s - b$ non-corner columns, so $\operatorname{per} M = \sum_{|T|=b} \operatorname{per}(B[\cdot, T]) \operatorname{per}([D \mid E[\cdot, C_2 \backslash T]])$, a sum of products of two *all-odd* permanents of sizes b and $s - b$; each all-odd $k \times k$ permanent has $v_2 \geq k - \lfloor \log_2 k \rfloor - 1$ (Theorem 6's engine: a permanent of an odd matrix $= J + 2E$ expands multilinearly as $\sum_T 2^{|T|} \operatorname{per}_T$ with $(k - |T|)! \mid \operatorname{per}_T$), so each summand has $v_2 \geq s - 2\lfloor \log_2 s \rfloor - 2$.

Therefore $v_2(\mathrm{Rest}_\rho) \geq (n - |\rho|) - 2\lfloor \log_2 n \rfloor - 2$ and $v_2(\mathrm{term}_\rho) \geq |\rho| + (n - |\rho|) - 2\lfloor \log_2 n \rfloor - 2 = n - 2\lfloor \log_2 n \rfloor - 2$; as $a(n)$ is their sum, $v_2(a(n)) \geq n - 2\lfloor \log_2 n \rfloor - 2$. □

Remark 10 (formalization — fully machine-checked). Theorem 9 is *formalized end to end in Lean 4/Mathlib*: `Paper3C1.two_pow_dvd_permanent_c1`: $2^{n - 2\lfloor \log_2 n \rfloor - 2} \mid$ (`gcdMat` n)`.permanent`, standard axioms `[propext, Classical.choice, Quot.sound]`, no `sorry`. The Lean proof is even *shorter* than the hand proof above: rather than reindex $a(n)$ by permutation grade, it observes that the gcd matrix *is* a zeroed-corner matrix—$\gcd(k+1, i+1)$ is even iff both $k+1, i+1$ are even, so $a(n) = \operatorname{per}\big(2\,ee + [\neg(k, i \text{ both even})]\big)$—and applies the reusable engine `ZeroedCorner.two_pow_dvd_permanent_zeroed`. That engine (column-multilinear expansion, each term carrying two groups of identical mask-columns) rests on a *two-group factorial* `TwoGroup.two_factorial_dvd_permanent` ($|A|!\,|B|! \mid \operatorname{per} M$ for two disjoint groups of identical rows, via a free $\mathrm{Sym}(A) \times \mathrm{Sym}(B)$ action—the product-group generalisation of the all-ones-rows kernel behind Theorem 6). The constant $c = 1$ is optimal *conditional* on the empirical upper bound $v_2(a(n)) \leq v_2(n!) + O(1) \sim n$ (observed through $n = 34$, unproven); what is established here is the lower bound, hence $\liminf_n v_2(a(n))/n \geq 1$.

3 Growth rate

Write $b(n) = a(n)/n! = \mathbb{E}_\sigma \prod_i \gcd(i, \sigma(i))$.

Theorem 11. $(a(n)/n!)^{1/n} = O\big((\log n)^{\log 2}\big)$*; equivalently* $a(n) \leq n!\,(\log n)^{(\log 2 + o(1))n}$.

Proof. $\operatorname{per}(M) \leq \prod_i R_i$ with $R_i = \sum_j \gcd(i, j) \leq n f(i)$, $f(i) = \sum_{d \mid i} \varphi(d)/d = \prod_{p^a \| i}(1 + a(1 - 1/p))$. By Stirling, $(a/n!)^{1/n} \leq e \exp\big(\frac{1}{n} \sum_{i \leq n} \log f(i)\big)$. Swapping sums, $\frac{1}{n} \sum_{i \leq n} \log f(i) = \sum_{p \leq n} g(p) + o(1)$ with $g(p) = \sum_{k \geq 1}(1 - \frac{1}{p}) p^{-k} \log(1 + k(1 - \frac{1}{p})) = \frac{\log 2}{p} + O(p^{-2})$. By Mertens, $\sum_{p \leq n} g(p) = (\log 2) \log\log n + O(1)$, giving the bound. □

Lemma 12 (entropy lower bound)**.** *For every doubly stochastic* $w = (w_{ij})$,

$$\frac{a(n)}{n!} \geq \frac{1}{n^n} \exp\Big(\sum_{ij} w_{ij} \log \gcd(i, j) + H(w)\Big), \qquad H(w) = -\sum_{ij} w_{ij} \log w_{ij}.$$

Proof. Let $\operatorname{cap}(M) = \inf_{\prod z_j = 1} \prod_i (Mz)_i$. Weighted AM–GM (using $\sum_j w_{ij} = 1$) gives $(Mz)_i = \sum_j M_{ij} z_j \geq \prod_j (M_{ij} z_j / w_{ij})^{w_{ij}}$, so, using $\sum_i w_{ij} = 1$ and $\prod_j z_j = 1$, $\prod_i (Mz)_i \geq \prod_{ij} (M_{ij}/w_{ij})^{w_{ij}} = \exp(\sum w_{ij} \log M_{ij} + H(w))$. Hence $\operatorname{cap}(M) \geq \exp(\sum w \log M + H(w))$, and Gurvits' theorem $\operatorname{per}(M) \geq (n!/n^n) \operatorname{cap}(M)$ (which for doubly stochastic matrices is exactly Van der Waerden–Falikman–Egorychev) finishes. □

Theorem 13 (growth rate)**.** $(a(n)/n!)^{1/n} \to \infty$. *Writing* $(a(n)/n!)^{1/n} = (\log n)^{\theta_n}$*, the exponent obeys* $\liminf_n \theta_n \geq 2 \ln \varphi - \varphi^{-2} = 0.5804\ldots$ *and* $\limsup_n \theta_n \leq \log 2$*; numerically* $\theta_n \approx 0.68$ *but in a strongly pre-asymptotic regime (*$\log\log n \approx 2$ *even at* $n = 3000$*), so the exact value—in particular whether* $\theta = \log 2$*—is open.*

Proof. Theorem 11 gives $\theta_n \le \log 2 + o(1)$; we prove the lower bound $\theta_n \ge 2\ln\varphi - \varphi^{-2} - o(1)$ (in particular $(a/n!)^{1/n} \to \infty$). Fix a cutoff P. Since $\prod_{p\le P,\, p|i,\, p|j} p$ divides $\gcd(i,j)$, monotonicity of the permanent gives $a(n) \ge \mathrm{per}(\tilde{M})$ with $\log \tilde{M}_{ij} = \sum_{p\le P} (\log p)\, \mathbf{1}[p \mid i]\, \mathbf{1}[p \mid j]$. Partition $[1,n]$ by divisibility pattern $S(i) = \{p \le P : p \mid i\}$: for $S \subseteq \{p \le P\}$, whenever $\prod_{p\le P} p \mid n$ the interval is a union of complete residue systems mod $\prod_{p\le P} p$, so

$$|V_S| = n\mu_S \ \textit{exactly}, \qquad \mu_S = \prod_{p\in S} \tfrac{1}{p} \prod_{p\notin S} \left(1 - \tfrac{1}{p}\right)$$

(for general n, $|V_S| = n\mu_S + O(2^{\pi(P)})$, the error constant in n). Let $\rho = \bigotimes_{p\le P} \rho^{(p)}$ be the product of the optimal Bernoulli couplings of Theorem 15, and set $w_{ij} = n\, \rho_{S(i)S(j)}/(|V_{S(i)}||V_{S(j)}|)$, constant on the $2^{\pi(P)} \times 2^{\pi(P)}$ blocks. When $\prod_{p\le P} p \mid n$ this w is *exactly* doubly stochastic, and a direct block computation gives

$$\tfrac{1}{n} F(w) - \log n = \sum_{p\le P} \Big[(\log p)\, \rho^{(p)}_{11} - D\big(\rho^{(p)} \,\|\, \mathrm{Bern}(\tfrac{1}{p})^{\otimes 2}\big) \Big] = \sum_{p\le P} g_p.$$

By Lemma 12, $(a(n)/n!)^{1/n} \ge e^{\sum_{p\le P} g_p}$ along $\prod_{p\le P} p \mid n$ (for general n a Sinkhorn correction on the $2^{\pi(P)}$ blocks changes F by $O_P(1) = o(n)$, giving the same limit); hence $\liminf_n (a(n)/n!)^{1/n} \ge e^{\sum_{p\le P} g_p}$ for *every* P. By Theorem 15 $\sum_{p\le P} g_p \to \infty$ as $P \to \infty$, so $(a(n)/n!)^{1/n} \to \infty$. □

Remark 14 (the exponent gap $[\,2\ln\varphi - \varphi^{-2},\ \log 2\,]$). Since $\mathrm{per}(M) \le \mathrm{cap}(M) \le \prod_i R_i$ and $\mathrm{per}(M) \ge (n!/n^n)\,\mathrm{cap}(M)$, one has $(a(n)/n!)^{1/n} = \Theta\big(e^{\max_w F(w)/n - \log n}\big)$ (constant in $[1,e]$), so the exponent is governed by the true capacity $\log\mathrm{cap}(M) = \max_w F(w)$. Our construction couples primes *independently*, reaching only $2\ln\varphi - \varphi^{-2}$; the bound $\theta \le \log 2$ uses $R_i = \sum_j \gcd(i,j)$, which feels all primes at once. Numerically the capacity exponent rises—$\theta_n = 0.672, 0.685, 0.689$ at $n = 10^2, 10^3, 3\cdot 10^3$, with $\log 2 - \theta_n = 0.021, 0.008, 0.004$ apparently $\to 0$—but this is unreliable: the *upper* rate $E_i[\log f(i)]/\log\log n$ itself converges to $\log 2$ only *logarithmically slowly* (still ≈ 0.74 at $n = 10^7$), so at any feasible size the data cannot separate $\theta = \log 2$ from a strictly smaller constant (the Sinkhorn deficit $\sum_i \log R_i - \log\mathrm{cap}(M)$ is $\approx 0.026\, n \log\log n$ at these n, consistent with either). The obstruction to a clean answer is the hard constraint $\sum_p v_p(i) \log p = \log i$, which couples all primes and defeats the independent-prime (product) analysis: the golden-ratio value $2\ln\varphi - \varphi^{-2}$ is the exponent of that *mean-field* strategy, cross-prime correlation provably beats it, and how far it goes—to $\log 2$ or short of it—is the open question.

Theorem 15 (per-prime gain; a golden-ratio constant)**.** *For a prime p define the divisibility gain*

$$g_p = \sup_{\rho} \Big[(\log p)\, \rho_{11} - D\big(\rho \,\|\, \mathrm{Bern}(\tfrac{1}{p})^{\otimes 2}\big) \Big],$$

the supremum over 2×2 couplings ρ with both marginals $\mathrm{Bern}(1/p)$*—the law of* $\mathbf{1}[p \mid i]$*—of the divisibility indicator with itself. Then $g_p > 0$ for every p, and*

$$\lim_{p\to\infty} p\, g_p = 2\ln\varphi - \varphi^{-2} = 2\ln\varphi + \varphi - 2 = 0.5804576\ldots, \qquad \varphi = \tfrac{1+\sqrt{5}}{2},$$

attained at $\rho_{11} = t^*/p$ with $t^* = \varphi^{-2} = \frac{3-\sqrt{5}}{2}$, the golden section (root of $(1-t)^2 = t$). Hence $\sum_p g_p = \infty$, with partial sums $\sim (2\ln\varphi - \varphi^{-2}) \log\log P$.

Proof. Write $\varepsilon = 1/p$ and $\rho_{11} = \varepsilon y$ ($y \in [0,1]$), so $\rho_{10} = \rho_{01} = \varepsilon(1-y)$ and $\rho_{00} = 1-2\varepsilon+\varepsilon y$. In the four relative-entropy terms the only unbounded piece, $\rho_{11}\log(\rho_{11}/\varepsilon^2) = \varepsilon y \log y + \varepsilon y \log p$, has its $\varepsilon y \log p$ cancelled *exactly* by the gain $(\log p)\rho_{11}$; the rest expands with an absolute $O(\varepsilon^2)$ error (no logarithm), giving

$$g_p(y) = \varepsilon\,\Psi(y) + O(\varepsilon^2), \qquad \Psi(y) = -y - y\log y - 2(1-y)\log(1-y).$$

As $p\varepsilon = 1$, $p\,g_p(y) = \Psi(y) + O(1/p)$; maximizing Ψ at $\Psi'(y) = \log\frac{(1-y)^2}{y} = 0$, i.e. $(1-y)^2 = y$ and $y = \varphi^{-2}$, yields $p\,g_p \geq \Psi(\varphi^{-2}) - O(1/p) = 2\ln\varphi - \varphi^{-2} - O(1/p)$, whence $\liminf_p p\,g_p \geq 2\ln\varphi - \varphi^{-2}$ (numerically an equality, $p\,g_p \uparrow 0.58046$). Positivity at finite p is the strict suboptimality of the independent coupling. The explicit $O(1/p)$ makes $g_p \geq (2\ln\varphi - \varphi^{-2} - O(1/p))/p$, so $\sum_p g_p = \infty$. ☐

Remark 16 (the finer coupling). Using the full valuation $v_p(i)$ in place of $\mathbf{1}[p \mid i]$ gives $G_p = \sup_\pi[(\log p)\mathbb{E}_\pi \min(a,b) - D(\pi \| q_p \otimes q_p)] \geq g_p$ ($q_{p,k} = (1-\frac{1}{p})p^{-k}$), with the clean identity $\mathbb{E}_{q_p\otimes q_p}[p^{\min}] = 1 + \frac{1}{p}$—so $G_p = \log(1+\frac{1}{p}) - \mathrm{SB}_p$, a Schrödinger-bridge cost—and the *same* limit $pG_p \to 2\ln\varphi - \varphi^{-2}$. The coarse divisibility gain g_p already suffices for Theorem 13. The divergence is genuinely *annealed* and multi-scale—invisible to the uniform measure (Jensen sees only 0.570), to any $O(1)$-tilt, and to any single block—and the golden section φ^{-2} is the signature of the entropy tradeoff Ψ.

4 The 2-adic deficit and two odd-permanent engines

Theorem 9 gives $v_2(a(n)) \geq n - 2\lfloor \log_2 n \rfloor - 2$, i.e. the deficit $D(n) := v_2(n!) - v_2(a(n))$ is $O(\log n)$. We now explain its *value*. Exact Gray-code Ryser data at the peaks $n = 2^k+1$ gives $v_2(a) = 3, 5, 11, 25$ for $k = 2,3,4,5$, which is

$$v_2\big(a(2^k+1)\big) = 2^k - 2k + 3, \qquad D(2^k+1) = 2k - 4 \sim 2\log_2 n \qquad (k = 2,3,4,5). \tag{1}$$

We stress that (1) is data, not a theorem: what follows explains those four values and proves the corresponding statement for the weight-zero grade N_0 of a, at every k with odd achiever parity. Whether $v_2(a) = v_2(N_0)$ persists beyond $k = 5$ we do not know. The mechanism behind (1) is two exact 2-adic valuation theorems for odd permanents. For an $s \times s$ all-odd matrix write $M = 2e + J$ (J all-ones); the column-multilinear expansion is $\mathrm{per}(2e+J) = \sum_{k=0}^{s} 2^k (s-k)!\,\pi_k(e)$, $\pi_k = \sum_{|R|=|S|=k} \mathrm{per}(e[R,S])$, so term k has $v_2 = s - s_2(s-k) + v_2(\pi_k)$ (with s_2 the binary digit sum).

Theorem 17 (A-engine)**.** *If $s = 2^a$, $a \geq 2$, and $\pi_1(e) = \sum_{i,j} e_{ij}$ is odd, then $v_2(\mathrm{per}(2e+J)) = 2^a - a$.*

Proof. The $k=1$ term has $v_2 = 2^a - s_2(2^a-1) + v_2(\pi_1) = 2^a - a$ (as π_1 is odd); the $k=0$ term has $v_2 = 2^a - 1 > 2^a - a$; and for $2 \leq k \leq 2^a$ every $2^a - k \in [0, 2^a-2]$ has $s_2 \leq a-1$ (only $2^a - 1$ has a ones), so those terms have $v_2 \geq 2^a - a + 1$. The $k=1$ term is the unique minimum: no cancellation, hence the claim. ☐

Theorem 18 (B-engine)**.** *If $s = 2^a+1$, $a \geq 3$, and $\pi_2(e)$ is odd, then $v_2(\mathrm{per}(2e+J)) = 2^a + 1 - a$.*

Proof. Now $\max_k s_2(s-k) = a$ is attained uniquely at $k = 2$ ($s-k = 2^a - 1$), giving that term $v_2 = 2^a + 1 - a + v_2(\pi_2) = 2^a + 1 - a$ (as π_2 is odd); all other k have $s_2(s-k) \le a-1$, hence $v_2 \ge 2^a + 2 - a$. Unique minimum, no cancellation. $\square$

Theorem 17 is the tight matching upper bound to Theorem 6, and it is **fully machine-checked**: the lower half $2^{2^a - a} \mid \operatorname{per}(2e + J)$ (`OddEngine.two_pow_dvd_permanent_odd_pow2`) and the upper half $2^{2^a - a + 1} \nmid \operatorname{per}(2e + J)$ under π_1 odd (`OddEngine.engine_upper`), together giving $v_2(\operatorname{per}(2e + J)) = 2^a - a$ exactly (Lean 4/Mathlib, standard axioms, no `sorry`). The formalized proof splits the column-subset expansion by $|t|$: the $|t|{=}0$ term is $\operatorname{per} J = s!$, the $|t|{\ge}2$ terms vanish mod $2^{2^a - a+1}$ (via the binary-complement identity $s_2(x) + s_2(2^a{-}1{-}x) = a$), and the $|t|{=}1$ terms sum to $2(s{-}1)!\,\pi_1$ (via the orbit–stabilizer fiber count $\#\{\sigma : \sigma j = r\} = (s{-}1)!$).

For odd n the weight-0 grade of $a(n) = \sum_w 2^w N_w$ factors as $N_0 = \sum_v \operatorname{per}(A_v)\operatorname{per}(B_v)$, where A_v (size $\lfloor n/2 \rfloor$) and B_v (size $\lfloor n/2 \rfloor + 1$) are all-odd. At the peaks $\lfloor n/2 \rfloor = 2^{k-1}$, so Theorems 17–18 apply and, when $\pi_1(A_v), \pi_2(B_v)$ are both odd,

$$v_2(\operatorname{per} A_v) + v_2(\operatorname{per} B_v) = \big(2^{k-1} - (k-1)\big) + \big(2^{k-1} + 1 - (k-1)\big) = 2^k - 2k + 3.$$

These are the minimum-valuation terms of N_0, so the leading-term rule gives

$$v_2(N_0) = 2^k - 2k + 3 \iff \#\{v : \pi_1(A_v), \pi_2(B_v) \text{ both odd}\} \text{ is odd},$$

reproducing $v_2(N_0) = 11, 25, 58$ at $k = 4, 5, 6$ exactly (the 58 is the even-parity lift off 55). This achiever count is computable in $O(n \log n)$ with no permanent (a divisor sieve); it is odd for $k \in \{4, 5, 7, 8, 11, 14, 16, 17, 18, 22, 23\}$ within $k \le 23$ (11 of 20, recurring), so $D_{N_0}(2^k + 1) = 2k - 4$ there, reaching 42 at $k = 23$ and setting a new record at every odd-parity peak.

Conjecture 19. *The achiever count is odd for infinitely many k; equivalently $D_{N_0}(2^k + 1)$ is unbounded, $= \Theta(\log n)$ along the peaks.*

For the full permanent one needs additionally $v_2(a) = v_2(N_0)$, which holds at $k = 2, 3, 4, 5$ (though the grade-margins $0, 1, 3, 1$ are small—a bare tie at $k = 2$—so this dominance is fragile).

An empirical refinement. Computing $C(k)$ to $k = 35$ and stratifying by $k \bmod 6$ reveals one residue class that is not pseudorandom: $C(k) = 1$ for *every* $k \equiv 5 \pmod 6$ tested ($k = 5, 11, 17, 23, 29, 35$, i.e. 6/6), while each of the other five classes is genuinely mixed. This suggests the sharper

Conjecture 20. $C(k) = 1$ *for all* $k \equiv 5 \pmod 6$*; equivalently* $D_{N_0}(2^k + 1) = 2k - 4$ *on this class. This would imply Conjecture 19.*

This is a lead, not a near-theorem: six independent structural approaches (dominant-stratum, achiever involution, small-prime CRT localization, period-6 recurrence, χ_4-reduction, and a MacWilliams [7] character-sum expansion) each terminate at the same parity barrier. The achiever counts on the class are $1, 211, 12659, \ldots$ (odd but large and growing), so the constancy is the parity of a genuine correlation, not a lone surviving term. Two facts survive as rigorous byproducts. First, a cleaner form of the target: the achiever set is exactly the intersection of *two affine $\mathbb{F}_2$ hyperplanes* in the $\equiv 3$ (4) prime-power divisor profile of v, $A = \{v \text{ odd} \le n :$

$R(v) = 1 \wedge G(v) = 1\}$ with both R, G affine-linear (the apparent quadratic B-side terms cancel since $[D \mid v][E \mid v] = [\mathrm{lcm}(D, E) \mid v]$). Second, C is a genuine *cross-stratum* quantity, not a localized invariant: at $k = 23$ the bilinear correlation term and its diagonal and off-diagonal parts all vanish while $C = 1$ is carried by the linear term—so no single stratum drives the class, and any "dominant term" argument fails at $k = 23$. The class is distinguished from the mixed class $k \equiv 1$ (6) only by $2^k \bmod 7$ (4 versus 2; both give $3 \parallel n$), but that digit acts through the full B-side correlation, not as a lever. Proving Conjecture 20 therefore appears to require the same correlation-parity control as Conjecture 19: it sharpens the target to a clean residue class and a cleaner (two-hyperplane) form, without lowering the barrier. The remaining analytic handle—the $\mathbb{F}_2$-Fourier expansion $|A| = \frac{1}{4}(N - S_R - S_G + S_{R\oplus G})$, whose cross-term $S_{R\oplus G} = \sum_{v \text{ odd} \le n} f(v)$ has f multiplicative—meets the same barrier: f is a genuinely k-dependent multiplicative ± 1 function (not a Dirichlet character: its $f(p) = -1$ density swings $23\% \to 79\% \to 19\%$ across consecutive k), so no closed form or class-number identity applies; and since $N \equiv 1$ (16), the criterion $|A|$ odd $\iff$ $N - S_R - S_G + S_{R\oplus G} \equiv 4$ (8) is *identical* to the conjecture, not a reduction. Two structural correlates of the class survive computation ($k \le 17$): $3 \mid S_{R\oplus G}$ (equivalently $S_G \equiv 1$, $S_{R\oplus G} \equiv 0 \bmod 3$, both constant on the class) on every class member and no non-member (9/9), and the flip-set orientation is uniformly sparse ($c_k = 0$) on $k \equiv 5$ (6) whereas $k \equiv 1$ (6) flips regime between members. Neither is a parity lever ($\bmod 3 \perp \bmod 8$), so the honest status is: *empirically true, analytically barriered.* The root obstruction is precise: $|A|$ is not the size of an affine $\mathbb{F}_2$-subspace (which would be a power of 2), because v ranges over *integers* $\le n$, not over $\mathbb{F}_2^{|P|}$—so $|A|$ is a *lattice-point count* in an affine variety, with $|A|/(N/4)$ wandering in $[0.24, 0.96]$ (measured $k \le 17$), the anti-correlation being exactly the growing character sums. The domain constraint breaks the $\mathbb{F}_2$-linear structure, which is why it cannot force the parity. None of the standard analytic techniques (exponential sums / circle method, automatic-sequence and sum-of-digits methods, Dirichlet-series / transfer-operator analysis of the doubling map, and the random-multiplicative model) applies: each controls magnitude or equidistribution, whereas the target is a 2-adic last bit. The parity of $|A|$ thus sits in the *Selberg parity-problem* family (two Liouville-type multiplicative ± 1 functions), entangled with a Gelfond-type digit correlation and modelled by random-multiplicative fair-coin parity—the canonical warning being $p(n) \bmod 2$, where an exact circle-method formula yields nothing and only modular/ℓ-adic methods (absent here) succeed. The conjecture is genuinely open, and expected to require a new exact-arithmetic, not analytic, input.

The arithmetic of the parity: χ_4 and the orders of 2. The parity in Conjecture 19 is not opaque—it has an exact arithmetic form that identifies precisely why it is hard. Since gcd of two odd numbers is odd and $(g-1)/2$ is odd iff $g \equiv 3$ (4), the parity $\pi_1(A_v) \bmod 2$ counts pairs whose gcd $\equiv 3$ (4):

$$\pi_1(A_v) \equiv G_3 + g_3(v) \pmod 2, \qquad g_3(v) := \#\{\, i \le 2^{k-1} : \gcd(i, v) \equiv 3 \ (4) \,\},$$

with G_3 the same count over all values. Writing $[\, n \equiv 3 \ (4) \,] = (1 - \chi_4(n))/2$ for the non-principal character χ_4 mod 4 and Dirichlet-inverting, only prime-power divisors survive modulo 2:

$$g_3(v) \ \equiv \sum_{\substack{q \equiv 3 \ (4) \text{ prime} \\ a \ge 1, \ q^a \mid v}} \left\lfloor \frac{2^{k-1}}{q^a} \right\rfloor \pmod 2, \tag{2}$$

and $\lfloor 2^{k-1}/q^a \rfloor \bmod 2$ is exactly the $(k-1)$-th *binary digit of* $1/q^a$, periodic in k with period $\mathrm{ord}_{q^a}(2)$. Identity (2) is in fact a *theorem*: for odd g, reducing the factorization mod 4 gives $g \equiv (-1)^{\sum_{q\equiv 3(4)} v_q(g)}$ (4), so $[\, g \equiv 3\ (4)\,]$ is the parity of $\sum_{q\equiv 3(4)} v_q(g) = \sum_{q^a} [q^a \mid g]$, and summing over entries yields (2); $\pi_2(B_v)$ obeys the same skeleton with different digit weights. Thus the achiever count $C(k) = \sum_v P_1(v)P_2(v)$ is a *correlation, over all v, of two parities built from the multiplicative orders* $\mathrm{ord}_q(2)$ *of the primes* $q \equiv 3$ (4) *up to* 2^k. As $k \to \infty$ new such primes continually enter, each contributing a bit governed by its own order; the empirical pseudorandomness of $C(k)$ mod 2 is the (conjectural) pseudorandomness of these orders. This places Conjecture 19 against *two* known walls: the distribution of $\mathrm{ord}_q(2)$ is Artin-primitive-root territory (unconditional only in part; sharp forms need GRH), and extracting a *parity* of a counting function from density information is the classical parity barrier. So the deficit's unboundedness is not merely "open"—it reduces to, and is obstructed by, open problems on the orders of 2.

A handle beyond the character-sum barrier: transcendence. The parity barrier is a barrier only in the analytic frame, where one controls the *size* $|\sum \chi|$ of a character sum but never the *parity* of the underlying count. Passing to the automata/transcendence frame removes exactly that blindness. The parity sequence $C(k)$ is a sequence over $\mathbb{F}_2$, and by the proven equivalence $D_{N_0}(2^k + 1) = 2k - 4 \iff C(k) = 1$ we have a clean dichotomy.

Theorem 21 (reformulation)**.** *Let $F(x) = \sum_k C(k)\, x^k \in \mathbb{F}_2[[x]]$. Then $D_{N_0}(2^k + 1)$ is unbounded iff $C(k)$ is not eventually 0 iff F is not a polynomial. Moreover each of the following implies unboundedness, in increasing strength: F is transcendental over $\mathbb{F}_2(x)$ $\Rightarrow$ C is not 2-automatic $\Rightarrow$ C is not eventually periodic $\Rightarrow$ unbounded.*

The middle link is Christol's theorem (C is 2-automatic iff F is algebraic over $\mathbb{F}_2(x)$); the outer links are that eventually-zero $\Rightarrow$ eventually-periodic $\Rightarrow$ automatic. Unlike a size bound, algebraicity *is* sensitive to parity, so this is a genuine escape route rather than a restatement. It also has an explicit pole obstruction. Since each word is periodic with period $m_v = \mathrm{ord}_v(2)$ once v enters at scale κ_v,

$$F(x) \;=\; \sum_v x^{\kappa_v} \frac{A_v(x)}{1 - x^{m_v}} \pmod 2, \qquad \deg A_v < m_v,$$

a legitimate power series (each coefficient is a finite sum). A rational function has finitely many poles, but the denominators place poles at m_v-th roots of unity with m_v unbounded; so F rational forces the combined residue at every primitive d-th root of unity to vanish for all large d. By Zsygmondy each order $d \neq 6$ carries a primitive prime $q_d \mid 2^d - 1$ injecting a pole at ζ_d with residue $\propto A_{q_d}(\zeta_d)$—and A_{q_d} is, by the Prime-Word Theorem below, the binary-expansion word of $1/q_d$. Rationality thus demands an explicit infinite family of cancellations among the reciprocal-digit words of Mersenne primitive primes—a concrete no-conspiracy condition, though not obviously unattainable, since the smallest prime of a given order need not be a Mersenne prime. A divisor-sieve from the linearity lemma computes $C(k)$ in $O(n \log\log n)$ (validated against the direct engine, $k \leq 22$), giving $C(k)$ for $k \leq 36$: density $\frac{18}{33}$ and a Berlekamp–Massey linear-complexity profile $1, 2, 5, 6, 10, 13, 16$ that climbs in plateau-then-jump fashion (it held at 13 for six terms, jumped to 16 at $k = 32$, and has held there since). These are *computed*, not proved: the full-$C(k)$ sieve is archived (`code/rust_grade/fastC.rs`), but the Berlekamp–Massey step is not archived in this release, so we draw no rigorous non-periodicity conclusion from the complexity profile—and 33 terms cannot by themselves separate a high-complexity rational F from a transcendental one. Crucially, either way unboundedness needs only that F is not a *polynomial*, i.e. $C(k)$ is not eventually zero—and the direct evidence for that is strong and independent of the transcendence question: $C(k) = 1$ at 15 of the 29 peaks $k \le 32$, so $D_{N_0}(2^k+1) = 2k-4$ there; the six largest such deficits are $40, 42, 44, 54, 56, 58$ at $k = 22, 23, 24, 29, 30, 31$, with $C(k) = 0$ in the gap $25 \le k \le 28$. What Theorem 21 contributes is not a resolution but a relocation: the parity in question is a coefficient of an explicit $\mathbb{F}_2$ power series, an object to which the 2-kernel and Mahler's method apply and to which the analytic parity barrier does not speak.

Adaptive peaks: weakening the conjecture. The parity barrier above blocks the *fixed* sequence $n = 2^k + 1$; it does not force us to use it. The engines generalize: if $s = 2^a + d$ with $\max_{0 \le y \le d} s_2(y) \le a-2$, then the unique maximizer of $s_2(s-t)$ is $s-t = 2^a - 1$ (any $x = 2^a + y$, $y \le d$, has $s_2 = 1 + s_2(y) \le a-1$), so π_{d+1} *odd forces* $v_2(\mathrm{per}(2e+J)) = s-a$ *exactly* (the case $d = 0, 1$ recovers Theorems 17–18; the side condition is sharp—$a = 3$, $d = 3$ fails via the tie $s_2(7) = 3$). Consequently, for *every* odd c with $d = (c-1)/2$ small and $k \ge 5$, the peak $n = 2^k + c$ has factor sizes $2^{k-1} + d$ and $2^{k-1} + d + 1$, both engine-covered, and

$$D_{N_0}(2^k + c) = 2k - 3 - s_2(c) \iff \#\{v : \pi_{d+1}(A_v),\ \pi_{d+2}(B_v) \text{ both odd}\} \text{ odd}.$$

Moreover *all* the parities $\pi_j \bmod 2$ are computable with no permanents: by the mod-4 factorization identity the χ_4-matrix decomposes over $\mathbb{F}_2$ as a sum of rank-one divisor indicators $e = \sum_{q^a} \mathbf{1}_{q^a|\cdot}\mathbf{1}_{q^a|\cdot}^{\mathsf{T}}$, and Cauchy–Binet plus Möbius inversion on the partition lattice (blocks of size ≥ 3 carry even $(|B|-1)!$) give

$$\pi_j(e) \equiv \sum_{|T|=j} a_T^{\mathrm{row}}\, a_T^{\mathrm{col}} \pmod 2, \qquad a_T = \sum_{\substack{\text{partitions of } T \\ \text{into singletons/pairs}}} \prod_B N(\mathrm{lcm}\, B),$$

with $N(D)$ a $\lfloor m/D \rfloor$-type multiple count—pure divisor combinatorics (validated against brute-force minors and against all fixed-c=1 counts). Computing the resulting *adaptive-peak table* for $k \le 10$, $c \le 7$: every k has an odd-parity entry with $c \le 3$; in particular the c=1 failures $k = 6, 9, 10$ are all repaired at $c = 3$. The case $(k, c) = (6, 3)$ is confirmed *out of sample* by an independent exact computation: the theory demands $D_{N_0}(67) = 2 \cdot 6 - 3 - s_2(3) = 7$, and the measured value is 7. The open step is thus weakened from a single-sequence parity to a window disjunction:

Conjecture 22 (window)**.** *There is C_0 (empirically $C_0 = 3$ suffices for all $k \le 10$) such that for infinitely many k some odd $c \le C_0$ has odd achiever parity. Equivalently $D_{N_0}(2^k + c) \ge 2k - 3 - s_2(c) \ge 2k - 5$ along a sequence of adaptive peaks, so the deficit of N_0 is unbounded.*

Conjecture 22 is still a parity statement—the barrier is weakened, not destroyed—but its failure would require the parity to vanish *simultaneously across the whole window* for all large k, a conspiracy across several independent-looking divisor correlations, and each instance is a cheap divisor computation.

The collapsible family. One further family exists on the other side: $n = 2^k - 1$, whose A-factor has size $2^{k-1} - 1$ and fires *unconditionally* (the extremal term is $t = 0$ with $\pi_0 = 1$). The achiever set is then cut out by the single condition $\pi_1(B_v)$ odd, and—unlike the two-condition families—its count parity *collapses*: the number of odd v is even, so the constant term drops, and swapping the v-sum to the divisor side gives (with $r_D = s_D$ here)

$$C_{-1}(k) \;\equiv\; \#\{\, q^a \le n,\ q \equiv 3\ (4)\ :\ \lfloor n/q^a \rfloor \equiv 1, 2 \pmod 4 \,\} \pmod 2,$$

the parity of an alternating count of prime powers in the harmonic intervals $\bigcup_j (n/(4j{+}3),\, n/(4j{+}1)]$. This parity *provably vanishes*:

Theorem 23 (the collapsible family never fires)**.** *For $k \ge 4$, the achiever count of the $n = 2^k - 1$ family is even; consequently $v_2(N_0(2^k - 1))$ always lifts strictly above the engine minimum.*

Proof. By the collapse, the count is $\equiv \sum_{D \in P} \mathrm{oddmult}(D) \pmod 2$, where $\mathrm{oddmult}(D) = \#\{x \text{ odd} \le n : D \mid x\}$. Swapping the order of summation (the divisor hyperbola) gives $\sum_{x \text{ odd} \le n} \Omega_3(x)$ with $\Omega_3(x) = \sum_{q \equiv 3(4)} v_q(x)$, and by the pointwise mod-4 identity $\Omega_3(x) \equiv [\, x \equiv 3\ (4)\,] \pmod 2$. Hence the count is $\equiv \#\{x \le 2^k - 1 : x \equiv 3\ (4)\} = 2^{k-2} \equiv 0 \pmod 2$. $\square$

(This matches the data: set sizes $4, 6, 10, 18, 24, 44, 76, 126, 226, 410, 744$ for $k = 4, \ldots, 14$, all even, and the measured massive lift $D_{N_0}(63) = 0$.) Theorem 23 sharpens the obstruction picture into a proved dichotomy: *linear* (single-condition) parities collapse via the hyperbola swap to χ_4-counts and vanish identically, so the deficit's unboundedness lives *exactly* in the non-collapsible correlation families of Conjecture 22.

The normal form: the window count is an $\mathbb{F}_2$ bilinear form. The correlation families themselves admit a drastic reduction. The key is a *linearity lemma*: shifting the column-count function by the divisor indicator $\chi_v(D) = [D \mid v]$ perturbs every matching sum only linearly mod 2,

$$\mathrm{ms}_j(T,\, N \pm \chi_v) \;\equiv\; \mathrm{ms}_j(T, N) \;+\; \sum_{D \in T} [D \mid v] \cdot \mathrm{ms}_{j-1}(T \setminus D,\, N) \pmod 2,$$

proved by induction on the matching recursion: the quadratic terms $[D \mid v][E \mid v]$ arising from the singleton product and from the pair block (via $[\mathrm{lcm}(D, E) \mid v] = [D \mid v][E \mid v]$) occur in equal pairs and cancel mod 2 (verified exhaustively for $j \le 4$). Hence *every* π_j-parity is affine-linear in the divisibility vector, $P(v) = b \oplus \sum_D \gamma_D [D \mid v]$ with computable $\gamma_D = \sum_{T \ni D} \mathrm{ms}_j(T, \mathrm{row})\, \mathrm{ms}_{j-1}(T \setminus D, \mathrm{col})$, and summing $P_1 P_2$ over v (hyperbola swap again) collapses the achiever count of *every* adaptive-peak family to

$$C(k, c) \;\equiv\; b_1 b_2 V \;\oplus\; b_2 L(\alpha) \;\oplus\; b_1 L(\beta) \;\oplus\; \alpha^{\mathsf{T}} M\, \beta \pmod 2, \tag{3}$$

with $V = \lceil (c{+}1)/2 \rceil \bmod 2$, $L(\gamma) = \sum_D \gamma_D\, \mathrm{oddmult}(D)$, and kernel matrix $M_{DE} = \mathrm{oddmult}(\mathrm{lcm}(D, E))$ over the prime powers $q^a \le n$, $q \equiv 3\ (4)$. Formula (3) is validated against all exactly-computed counts (13 families, plus cross-engine checks). Conjecture 22 is thereby a statement about a single explicit $\mathbb{F}_2$ *bilinear form* in two computable divisor vectors—the v-sum is gone—and the per-family computation drops to one subset enumeration plus $O(|P|^2)$, eliminating the per-v phase entirely.

Three structural facts about this form (computed $k \leq 9$): M is nearly full-rank over $\mathbb{F}_2$ (corank ≤ 5 at $|P| = 35$), so no kernel degeneracy helps; the B-side support S_β is *sparse and nearly constant* ($|S_\beta| \approx 7$ while $|P|$ grows), so $\alpha^\mathsf{T} M\beta$ is a XOR of only $\sim$7 twisted linear sums; and the A-side linear term reduces exactly (verified $k \leq 12$) to a *digit–remainder* prime sum: with $q = \lfloor 2^{k-1}/D \rfloor$, $\rho = 2^{k-1} \bmod D$,

$$L(\alpha) \ \equiv \ \#\{\, D \in P : \ q \text{ odd}, \ \rho < (D{-}1)/2 \,\} \pmod 2.$$

At the count level such sums are classical equidistribution of fractional parts N/p; at the parity level they are Gelfond-type digit-conditions-over-primes statements—the problem family where count-level results were achieved by Mauduit–Rivat-style exponential-sum methods. Equivalently (an identity we verify exactly), $L(\alpha)$ is the parity of $\#\{D \in P : 2^{k-1} \bmod 2D \in [D, 3D/2)\}$: where the *orbit of* 2 lands in $\mathbb{Z}/2D$.

Moreover the B-side support is *band-structured*: β_E is constant on every band $\{E : \lfloor n/E \rfloor = t\}$ with $t \geq 3$ (verified $k \leq 10$; the exceptions at $t \leq 2$ are tower-driven), so both α and β are interval-indicator vectors up to bounded corrections. The bilinear term then reduces to a sum over band-pairs of parities of counts of *products* $DE \leq n$ of prime powers $\equiv 3$ (4) in prescribed intervals. The window conjecture is thereby a bounded combination of parities of prime- and semiprime-counting functions in explicit intervals. This is the endpoint of the reduction, and it is two-sided: parity of prime-counting functions is itself a famous open problem (the barrier in its most classical dress), yet Theorem 23 shows such combinations *can* telescope to provable arithmetic-progression counts.

Telescope versus residue: the verdict. Pushing the swap calculus fully (each step verified exactly, $k \leq 13$): every *pure digit sum* telescopes and is provable—e.g. $\sum_{D \in P} \lfloor 2^{k-1}/D \rfloor \bmod 2 \equiv \#\{v \leq 2^{k-1} : \mathrm{oddpart}(v) \equiv 3 \ (4)\} = 2^{k-2} - 1 \equiv 1$, by the same swap-plus-pointwise-χ_4 mechanism as Theorem 23—and the c=1 linear term decomposes exactly as

$$L(\alpha) \ = \ 1 \oplus \underbrace{\#\{D \in P : \text{binary digits } (k{-}1, k) \text{ of } 1/D \text{ both } 1\}}_{\mathrm{Corr}(k)} \oplus \underbrace{\#\{D \in P : D \mid 2^k{+}1, \ \lfloor 2^{k-1}/D \rfloor \text{ odd}\}}_{\mathrm{Div}(k)},$$

with Div an explicit factorization term. What resists is precise: single digits telescope, digit *pairs* do not (Corr is the adjacent-digit correlation of $1/D$ over $D \in P$); and for the bilinear term the u-swap gives (verified) $\alpha^\mathsf{T} M\beta \equiv \#\{u \text{ odd} \leq n : \sum_{D|u} \alpha_D \text{ odd} \ \wedge \ \sum_{E|u} \beta_E \text{ odd}\}$—*again* a correlation of two divisor-sum parities, the same shape as the original achiever count. The swap calculus thus strips all linear content to provable constants and factorization terms while **fixing the correlation shape**: correlations reproduce correlations under every available identity. The residue of Conjecture 22 is a fixed point of the reduction calculus—pairwise digit correlations of reciprocals of primes $\equiv 3$ (4)—so progress requires genuinely second-order input (joint digit equidistribution over primes at the parity level) or an identity outside the swap calculus. This, and not any single computation, is the precise remaining distance to unboundedness.

Two such outside identities exist, and they push the residue one level further (each verified exactly). First, the digit *product* is itself swappable—$q_{k-1}(D)q_k(D)$ counts pairs—so $\mathrm{Corr}(k) \equiv \#\{(v, w) : v \leq 2^{k-1}, w \leq 2^k, \ \mathrm{oddpart}(\gcd(v, w)) \equiv 3 \ (4)\} \pmod 2$, a two-dimensional gcd-$\chi_4$ lattice count. Second, on *square* domains the transposition involution $(v, w) \mapsto (w, v)$ cancels all off-diagonal pairs mod 2, so with $\kappa(a, b) := \#\{v \text{ odd} \leq 2^a, \ w \text{ odd} \leq$

$2^b : \gcd \equiv 3\ (4)\}$ mod 2 one gets the proven evaluation $\kappa(a,a) \equiv [a{=}2]$. A parity-quadrant recursion ($\gcd(2v,w) = \gcd(v,w)$ for odd w) reduces $\mathrm{Corr}(k)$ entirely to the dyadic table $\kappa(i,j)$, whose rows are eventually periodic and whose diagonal is provable; what remains pseudorandom is exactly the *near-diagonal strips* $\kappa(a, a{+}\delta)$, δ small (e.g. $\kappa(a, a{+}1) = 0,0,1,1,1,0,1,1,1$ for $a = 3,\dots,11$). Each identity outside the calculus has evaluated another stratum; the surviving core is the joint distribution of gcd-χ_4 counts across *adjacent dyadic scales*. Both provable parity strata (Theorem 23 and the digit-sum constant) are machine-checked (`Paper3CMinus1.sum_omega3_odd_even`, `Paper3DigitSum.sum_omega3_digit_odd`), as is the involution evaluation $\kappa(a,a) \equiv [a{=}2]$ (`Paper3Involution.kappa_diag`).

The full recursion *assembles*: chaining product-swap, quadrant splits, the involution, dyadic strips, and the translation identity yields an exact prime-free evaluator of $\mathrm{Corr}(k)$ in which every step is a proven identity (`code/hyperbola_assembly.py`; validated against brute lattice counts and against the prime-side definition, 17/17). Instrumenting the strata shows the involution contributions cancel globally, leaving exactly $\mathrm{Corr}(k) = \mathrm{FP}(k) \oplus \mathrm{BD}(k)$: the full-period stratum (the τ_3/hyperbola machinery above) and the boundary stratum, whose normal form is a sum of gcd-χ_4 counts over *doubling windows* $(r_v, 2r_v]$ mod $2v$ anchored at $r_v = 2^{j-1}$ mod $2v$. The remaining open core of Conjecture 22's linear stratum is thus a single explicitly-structured object: doubling-window gcd counts, a natural target for dynamical methods.

The dynamical analysis terminates in a *word normal form* (each step verified exactly). For fixed odd v the boundary bit is periodic in the scale i with period $\mathrm{ord}_v(2)$, so each v carries a fixed binary word; the word is zero *iff* v has no prime factor $\equiv 3\ (4)$ (both directions machine-checked: `Paper3CMinus1.gcd_chi4_empty_of_no_three` and `_nonempty_of_three`); and since $\{v : \mathrm{ord}_v(2) = m\}$ consists of divisors of $2^m - 1$,

$$\mathrm{BD}(i, i{+}1) \;=\; \bigoplus_{v\ \mathrm{odd} \le 2^i} \mathrm{word}_v\big[i \bmod \mathrm{ord}_v(2)\big]$$

is exactly a XOR of periodic itinerary words attached to divisors of Mersenne numbers, evaluated at the diagonal phase (a further reflection identity, $c_v(I) = c_v(-I)$ via $w \mapsto 2v - w$, constrains the words). The periodicity itself—that any two length-$2v$ windows carry the same gcd-χ_4 count, so the word is well-defined—is machine-checked (`Paper3Translation.count_period_eq`, via the bijection $w \mapsto w + 2vk$), as is its reflection companion. No further identity among swap, transposition, translation, and negation reduces the count; the words are the irreducible objects. The linear stratum of Conjecture 22 thus rests, in final form, on the joint irregularity of Mersenne factorizations and their divisors' gcd-χ_4 itinerary words: unproved, but fully mechanized, and any structural word-sum invariant would propagate up the proven chain to the deficit itself.

Word theory: invariants, sub-periods, and class numbers. The words admit their own theory (each statement verified exactly). *Word-sum invariant*: telescoping the cumulative count along the doubling orbit gives $\sum_i \mathrm{word}_v[i] \equiv \mathrm{wt}(v) \cdot t_3(v) \pmod 2$, where $\mathrm{wt}(v) = \#\{s \in \langle 2 \rangle \subseteq (\mathbb{Z}/v)^\times : s > v/2\}$ is the number of 1s in the binary period of $1/v$. This invariant is now MACHINE-CHECKED (`Paper3Translation.word_sum_invariant`, for odd v with $2^m \equiv 1\,(v)$ and scale ≥ 2), and unconditionally in m via Euler with $m = \varphi(v)$ (`word_sum_invariant_totient`). *Anti-period law*: if $2^{m/2} \equiv -1\ (v)$ then $\mathrm{word}_v[i + m/2] = \mathrm{word}_v[i] \oplus t_3(v)$ (negation identity plus even anchors; 28/28), which with its CRT partial-negation variant explains every observed sub-period: sub-period iff $t_3(v)$ even, complement-word iff odd. The single-period count $t_3(v)$ has itself a clean parity, machine-checked via the reflection involution: for odd v, $t_3(v) \equiv [v \equiv 3\ (4)] \pmod 2$ (`Paper3Negation.t3_card_parity`; the map $w \mapsto 2v-w$ is a gcd-preserving involution of the period whose unique fixed point $w = v$ contributes iff $v \equiv 3$ (4)). So sub-period $\iff$ $v \equiv 1$ (4), complement-word $\iff$ $v \equiv 3$ (4). *Weight evaluation*: for full-orbit primes $\mathrm{wt}(p) = (p-1)/2$, so every prime $p \equiv 3$ (4) with 2 primitive root carries an odd-sum word; and for *half-orbit* primes $p \equiv 3$ (4) (2 a quadratic residue, ord $= (p-1)/2$), the classical residue-distribution formula for $p \equiv 3 \pmod 4$ — the count of quadratic residues in $(0, p/2)$ exceeds the expected $(p-1)/4$ by $h(-p)/2$, a standard consequence of the class number formula [5, §6] — gives

$$\sum_i \mathrm{word}_p[i] \;\equiv\; \mathrm{wt}(p) \;=\; \tfrac{(p-1)/2-h(-p)}{2} \pmod 2,$$

verified for all thirteen such $p \le 359$: **the class number of $\mathbb{Q}(\sqrt{-p})$ governs the word-sum invariant**. The deficit mechanism of the gcd permanent is thus, in part, class-number-driven — the chain *deficit* $\to$ *achiever parity* $\to$ *normal form* $\to$ *Mersenne words* $\to$ *class numbers* being verified at every link and machine-checked at five.

The Prime-Word Theorem and n-gram class-number formulas. The words themselves are classical objects in disguise. For prime p: $\gcd(p, w) \equiv 3$ (4) iff $p \mid w$ (and $p \equiv 3$ (4)), each period contains exactly one odd multiple of p, and window membership reduces to (verified over all tested primes and phases)

$$\mathrm{word}_p[i] \;=\; \left[\, 2^i \bmod p > p/2 \,\right] \;=\; \text{the } i\text{-th binary digit of } 1/p.$$

The deficit boundary words of primes are the binary expansions of reciprocal primes; composite words decompose χ_4-linearly into prime-power multiple-itineraries. Consequently the k-gram statistics of a prime word are counts of the orbit of 2 in dyadic arcs; for half-orbit primes ($p \equiv 7 \bmod 8$, orbit = quadratic residues) these are *exact class-number formulas*, verified exhaustively: the digit-pair profile per period is

$$\#00 = \tfrac{p+4h-3}{8}, \quad \#01 = \#10 = \tfrac{p+1}{8}, \quad \#11 = \tfrac{p-4h-3}{8}, \qquad h = h(-p),$$

and the digit-triple profile (eighth arcs) involves both $h(-p)$ and $h(-8p)$: $B_0 = \frac{p-7+8h(-p)-2h(-8p)}{16}$, $B_1 = B_2 = B_4 = \frac{p+1+2h(-8p)}{16}$, $B_3 = B_5 = B_6 = \frac{p+1-2h(-8p)}{16}$, $B_7 = \frac{p-7-8h(-p)+2h(-8p)}{16}$ (adjacent pairs re-sum to the quarter formulas). The negation identity underlying the anti-period law is machine-checked (`Paper3Negation.count_reflect`).

Theorem 24 (the class-number ladder has depth exactly three)**.** *For $p \equiv 7 \pmod 8$ write $S_k(p) = \#\{\mathrm{QR} \text{ in } (0, p/2^k)\}$. Then*

$$S_1 = \tfrac{p-1+2h(-p)}{4}, \quad S_2 = \tfrac{p-3+4h(-p)}{8}, \quad S_3 = \tfrac{p-7+8h(-p)-2h(-8p)}{16},$$

while for $k \ge 4$ no expression $S_k = (p + a + b\,h(-p) + c\,h(-8p) + d\,h(-16p))/2^{k+1}$ with integers a, b, c, d exists. Consequently the density, digit-pair, and digit-triple word statistics are exact class-number formulas, and the digit-quadruple-and-finer statistics are not.

The obstruction is structural, not computational. A level-k evaluation requires a real primitive Dirichlet character of conductor 2^k to resolve the 2^k-division of the arc; but $(\mathbb{Z}/16)^\times \cong C_2 \times C_4$ has exactly four real characters, *all* of conductor ≤ 8 ($\mathbf{1},\ \chi_{-4},\ \chi_8,\ \chi_{-8}$), and there is no real primitive character of any conductor 2^k with $k \ge 4$. Equivalently, the only fundamental discriminants in the family $\chi_{-p}\cdot$(2-power) are $-p$ and $-8p$ ($-4p$ and $-16p$ being non-fundamental $2^{2j} \cdot (-p)$), so $h(-p)$ and $h(-8p)$ are the only class numbers available; they suffice through $k = 3$ and nothing reaches $k = 4$. Beyond the ladder the residual $|2^{k+1}S_k - p|$ is $O(\sqrt{p}\log p)$ (Pólya–Vinogradov), genuine equidistribution with no closed form. *The arithmetic content of the deficit is thus completely stratified: a finite class-number core of depth three over* $\mathbb{Q}(\sqrt{-p})$ *and* $\mathbb{Q}(\sqrt{-2p})$*, and a character-sum tail.*

A third identity type—*translation*, $w \mapsto w \pm 2v$ preserving $\gcd(v, w)$—splits the strip into full periods plus boundary windows, and the full-period part telescopes (each step verified exactly). Its foundation—that the gcd-χ_4 count over *any* length-$2v$ window is the same constant $t_3(v)$, so the counting function increments by exactly $t_3(v)$ per period—is machine-checked (`Paper3Translation.count_window_offset`, from the indicator's $2v$-periodicity via the shift and reflection identities); together with the period-block count `count_qperiods` (q periods $=$ $q \cdot t_3$) and the strip decomposition `count_strip_decomp` it forms the machine-checked telescoping core of the word-sum invariant; the range half—that the per-scale strip counts over $(2^i, 2^{i+1}]$ sum to the single count over $(1, 2^m]$—is also machine-checked (`count_strip_telescope`), as is the per-scale strip decomposition and its sum `boundary_sum` ($\sum$ strip $= t_3 \cdot \sum q + \sum$ boundary). These reduce the word-sum invariant, machine-checked, to $\sum_i \mathrm{word}_v[i] \equiv (\sum \mathrm{strip}) \oplus t_3 \cdot (\sum q)$ (mod 2); the final orbit step is assembled from $\sum \mathrm{strip} \equiv 0$ (`count_full_range_even`: the range spans an even number of periods, via $2^m \equiv 1\,(v)$) and $\sum q \equiv \mathrm{wt}(v)$ (the period-sum of $\lfloor 2^i/2v \rfloor$ is the digit-sum of $1/v$), giving the fully machine-checked $\sum \mathrm{word} \equiv \mathrm{wt} \cdot t_3$. Concretely: with $t = \chi_3 * \varphi$ one has $\mathrm{FP}(a) \equiv \sum_{u \le 2^{a-1}} \tau_3(\mathrm{oddpart}\, u) \pmod 2$, the pairing $d \mapsto n/d$ evaluates τ_3 mod 2 (a square-indicator stratum for $n \equiv 1$, and $\tau(n)/2$ for $n \equiv 3$), and exactly $\sum_{n \le X,\, n \equiv 3(4)} \tau(n)/2 = \#\{(d, e) : d \equiv 1,\ e \equiv 3\ (4),\ de \le X\}$—a two-AP hyperbola count, which Dirichlet's symmetry splits into $\sqrt{X}$-range sums (a range-shrinking recursion). The decisive structural consequence: **after the product-swap the residue is prime-free**—all remaining objects are lattice counts over the integers, where the classical hyperbola toolkit applies. The obstruction is thereby transformed from Gelfond-class digit parity over primes into (open, but mechanical-looking) hyperbola-recursion bookkeeping; completing that recursion, including the boundary strata, is the remaining distance to evaluating $\mathrm{Corr}(k)$ in closed form.

References

[1] H.J.S. Smith, *On the value of a certain arithmetical determinant*, Proc. London Math. Soc. (1876).

[2] L. Valiant, *The complexity of computing the permanent*, Theoret. Comput. Sci. **8** (1979).

[3] G.P. Egorychev, *The solution of van der Waerden's problem for permanents*, Adv. Math. **42** (1981); D.I. Falikman (1981).

[4] H.J. Ryser, *Combinatorial Mathematics*, Carus Mathematical Monographs **14**, MAA, 1963.

[5] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Graduate Texts in Mathematics **74**, Springer, 2000. (Class number formula and the distribution of quadratic residues; Pólya–Vinogradov.)

[6] G.H. Hardy, E.M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford, 2008. (Mertens' theorems.)

[7] F.J. MacWilliams, N.J.A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland, 1977.